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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 11/30/2018 |
| **PI Name:** | Paul, Anna-Lisa Ph.D. | | |
| **Project Title:** | Epigenetic Change in Arabidopsis thaliana in Response to Spaceflight - Differential Cytosine DNA Methylation of Plants on the ISS | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Reproductive Biology<br>(2) Developmental Biology | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 32611-0690 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2014 Space Biology Flight NNH14ZTT001N |
| **Start Date:** | 09/12/2014 | **End Date:** | 09/10/2018 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Levine, Howard | **Contact Phone:** | 321-861-3502 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: Extended to 9/10/2018 per NSSC information (Ed., 10/10/17) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Ferl, Robert Ph.D. ( University of Florida ) | | |
| **Grant/Contract No.:** | NNX14AT24G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | This project seeks to use an unbiased genome-wide approach to define the changes in DNA methylation that occur in Arabidopsis during spaceflight, in order to better understand the spectrum of responses that allow adaptation to the spaceflight environment. This project is one of both discovery on a genome-wide scale and application to specific hypotheses. We propose replicated whole genome bisulfite sequencing in Arabidopsis roots and leaves to discover and quantify 5-methylCytosine (5mCyt) epigenetic modification, coupled with RNASeq examination of the comparable transcriptomes to both quantitatively and qualitatively assess spaceflight-associated DNA methylation on transcript profiles. We will then test the hypothesis that methylation is involved in spaceflight adaptation by similarly examining methylation mutants.<br>Epigenetic features play a role in the regulation of gene expression and the subsequent response of an organism to its environment. DNA methylation is a major epigenetic modification that is not directly coded into the genome sequence and yet can modify expression and be inherited for at least for one generation. Several groups, including ours, have shown that large numbers of plant genes are differentially expressed in response to spaceflight.<br><br>We now have extensive preliminary data that show significant genome-wide 5mCyt modification differences in spaceflight plants compared to ground controls.<br><br>The tactical objectives of this proposal are: 1) Bisulfite whole genome sequencing will be used to generate maps of spaceflight-induced changes in 5mCyt at single base-pair resolution for roots and leaves. 2) RNASeq will be used to digitally quantify and qualitatively assess transcriptome in the same samples, as differential methylation has been functionally linked to changes in both promoter regulation and structural changes in RNAs. These data will characterize functional impacts of spaceflight induced across the entire genome. 3) DNA methylation mutants will be used to directly test the hypothesis that epigenetic modification of DNA is a key aspect of spaceflight adaptation.<br><br>Elucidating the spaceflight methylomes of plants contributes to the fundamental understanding of how biology responds and adapts to the spaceflight environment, and also provides insight in the adaptive strategies plants use in the face of novel stimuli outside of their evolutionary experience. The long term goal of this line of inquiry is to understand how the methylome contributes to the adaptive strategies of plants to novel environments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research contributes to a greater understanding of the role of epigenetic changes to the genome in response to novel or stressful environments. |
| **Task Progress:** | In this final year of the grant, the returned plant samples were extracted and evaluated, and then ensuing molecular data analyzed. So far, these data have contributed to two additional manuscripts; one in press, and one in the final stages of preparation. A synopsis of the results and conclusions follow:<br>Harvests<br><br>Plants for SVT (Science Verification Test) and EVT (Experiment Verification Test) were grown in VEGGIE and were harvested at 11 days. Each plate was harvested to a single Falcon tube containing RNA later. Samples were then stowed at -80Deg Celsius until returned to the Principal Investigators (PIs). Harvest from station was monitored by the PI via video communication from their lab in Gainesville, FL.<br><br>Post-Harvest Analysis<br><br>Samples for both flight and ground control were transferred by the PI to the University of Florida for analysis of quality and quantity. Samples were warmed and separated into root and leaf tissue, hypocotyl tissue was removed and stored for future analysis. Plates were then separated into 4 replicates, each having two plates within a replicate. A portion of the plants (2-4) were removed from each plate for RNA extraction, and then depending on genotype, 4-20 plants that were left on the plate were used for DNA extraction.<br><br>Extraction of Nucleic Acids from Flight and Ground Control Samples<br><br>RNA extraction was completed using a Qiagen RNeasy Plant kit. DNA extraction was preformed using a method developed in-house for high levels of clean genomic DNA. All sample material provided enough nucleic acid material for downstream applications.<br><br>Imaging and plant growth: Images of the first plates in VEGGIE were taken on Day 4, Day 8, and Day 11 (harvest). Due to excellent germination and growth all plates appear to have consistent growth. Photography of plates was sufficient for post-harvest analysis to provide data fro growth rates and morphometric analyses.<br><br>Methylome Analyses: To examine the role of 5mCyt in spaceflight adaptation, the APEX04-EPEX experiment conducted onboard the ISS evaluated the spaceflight altered genome-wide methylation profiles of two methylation-regulating gene mutants (methyltransferase 1 (met1-7) and elongator complex subunit 2 (elp2-5)) along with a wild type Col-0 control. The methylome data revealed that both elp2-5 and met1-7 mutants had, on average, significantly higher methylation levels in flight than in the ground controls for both leaf and root tissues, whereas there were no differences in methylation levels between flight samples and ground controls in Col-0 roots and leaves. Overlapping the differential gene expression data with differentially methylated cytosines within protein-coding gene regions revealed the complex transcriptional regulatory processes of adapting to spaceflight. The methylation mutants provided new insights into organ-specific adaptation to spaceflight. The results show that elp2-5 has a role in regulating root adaptation to spaceflight, whereas met1-7 has a more prominent role in the physiological adaptation of leaves. Gene ontology enrichment showed that a significant proportion of Col-0 differentially-methylated and expressed genes were associated with pathogen response consistent with previous reports. Met1-7 and elp2-5 showed enrichments unique to each organ and genotype. The major points of discovery were:<br><br>- Spaceflight increases genome-wide DNA methylation levels in met1-7 and elp2-5 but not in the wild type Col-0 plants<br><br>Average genome wide methylation level profiles in spaceflight and on the ground.<br><br>- Spaceflight changes in DNA methylation levels were associated with protein-coding gene regions.<br><br>- Spaceflight alters the distribution and direction of differential methylated cytosines (DmCs) and differential |

methylated regions (DMRs) in elp2-5 and met1-7 leaves and roots.

- There is a positive correlation of differential DNA methylation and differential gene expression. - The patterns of differential DNA methylation and differential gene expression in response to spaceflight differs among the methylation mutants and the wild type.

General conclusions

A growing number of studies are showing evidence that DNA methylation plays a crucial role in regulating stress responses and adaptation in plants. Understanding how plant systems adapt to novel environments, such as spaceflight, is important for advancing research in plant abiotic stress tolerance, which has valuable implications for ensuring food security in adverse environments. Previous spaceflight transcriptomic and proteomic data showed prominent spaceflight changes in genes typically associated with abiotic stress response on Earth. Therefore, to understand how plants adapt to the novel environment of spaceflight, it is important to investigate the epigenetic regulation of plants' genes in spaceflight and identify specific marks that contribute to known gene expression patterns.

In this study, DNA methylation dynamics in the leaves and roots of wild-type Col-0 and two methylation mutant (elp2 and met1) Arabidopsis lines grown in spaceflight and on the ground were assessed with a single-base resolution. The genome-wide analyses presented here show that the mechanisms that regulate spaceflight responses also involve changes in DNA methylation status in an organ-specific manner. Although spaceflight did not induce large scale genome-wide changes in wild type Col-0 leaves and roots, spaceflight did induce significant genome-wide methylation changes in the roots and leaves of both elp2-5 and met1-7.

Several observations reported here suggest that differential DNA methylation in response to spaceflight is highly organ-specific and affects a different set of genes, and the GO-enriched categories differed between leaves and roots in both the methylation mutants. Thus, the finding that global methylation patterns are drastically remodeled by spaceflight in an organ-specific manner could provide insight into how epigenetic marks influence plant genomes and transcriptional programs to respond and adapt to novel conditions.

| **Bibliography Type:** | Description: (Last Updated: 11/20/2025) |
|---|---|
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